Binary Tree Data Structure

What will be covered

- Concept Of Tree.
- General Tree strucure.
- Element of Tree.
- Binary Tree.
- Benifits of Binary Tree.
- Types Of Binary Tree
- Binary Tree Respresentation.

Concept Of Tree.

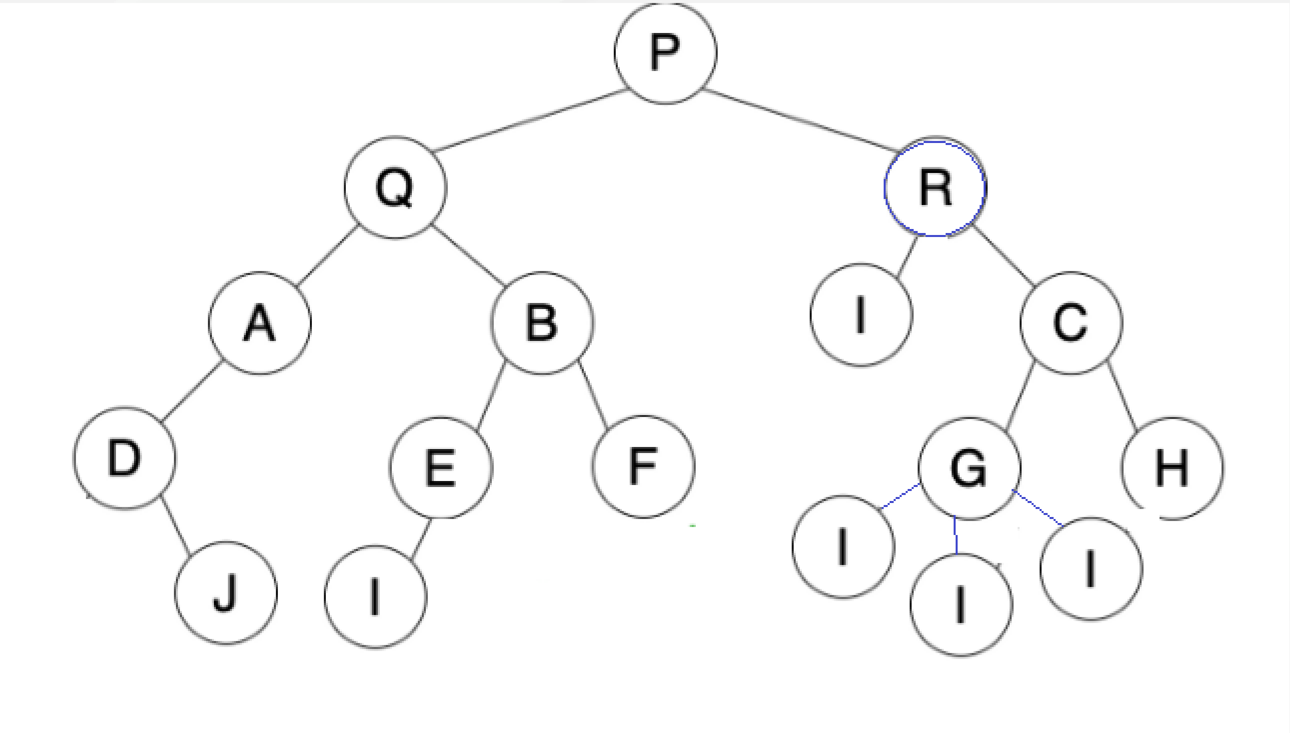
- A Tree is a Hierarchical data structure that naturally hierarchically stores the information. The Tree data structure is one of the most efficient and mature. The nodes connected by the edges are represented.

- Properties of Tree: Every tree has a specific root node. A root node can cross each tree node. It is called root, as the tree was the only root. Every child has only one parent, but the parent can have many children.

General Tree Data Structure

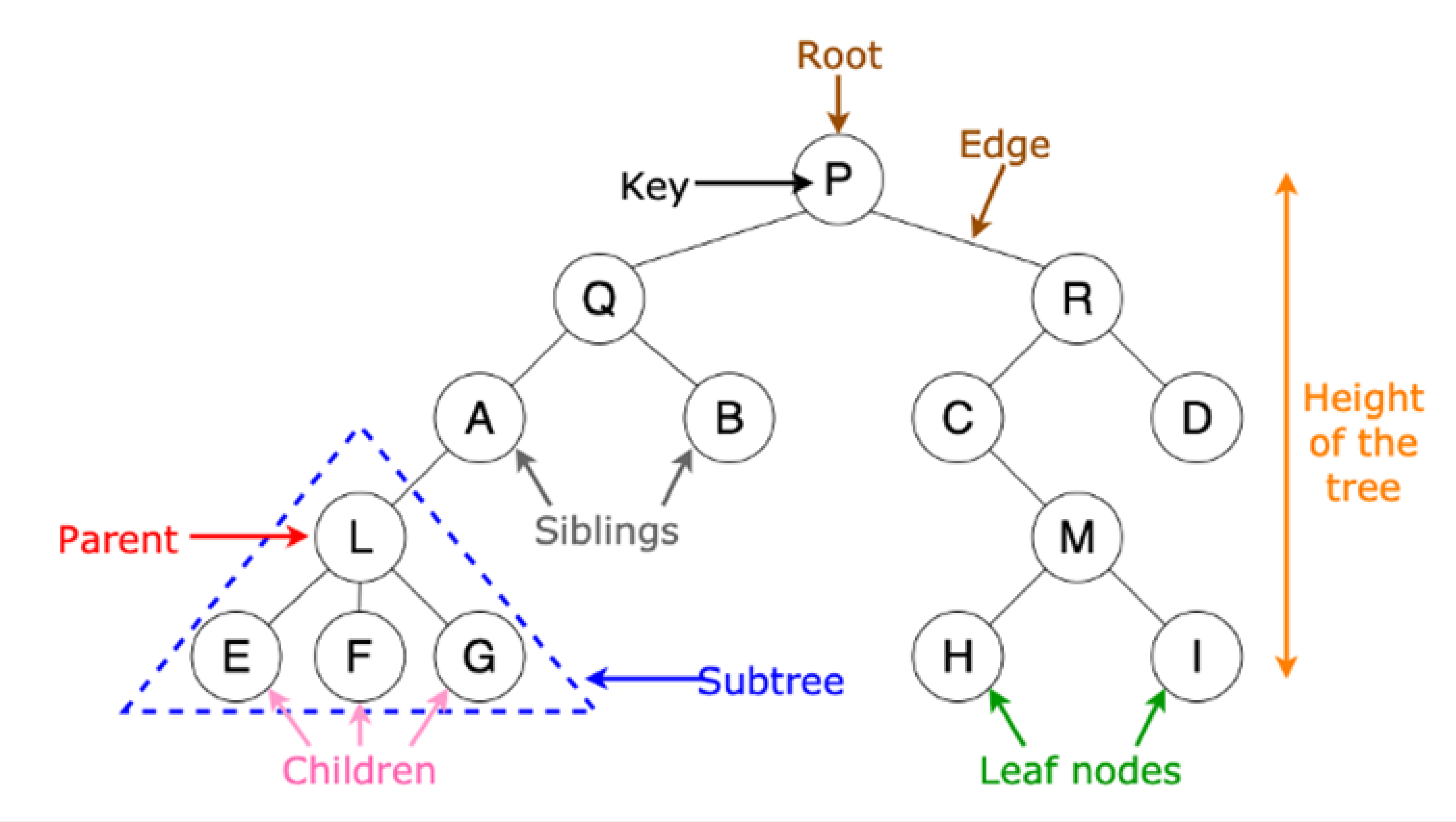
A general tree is a tree data structure where there are no constraints on the hierarchical structure.

Node can any number of children.



Elements Of Tree

1. Root
2. Key
3. Parent
4. Children
5. Subtree
6. Siblings
7. Root
8. Leaf Nodes
9. Height of the tree

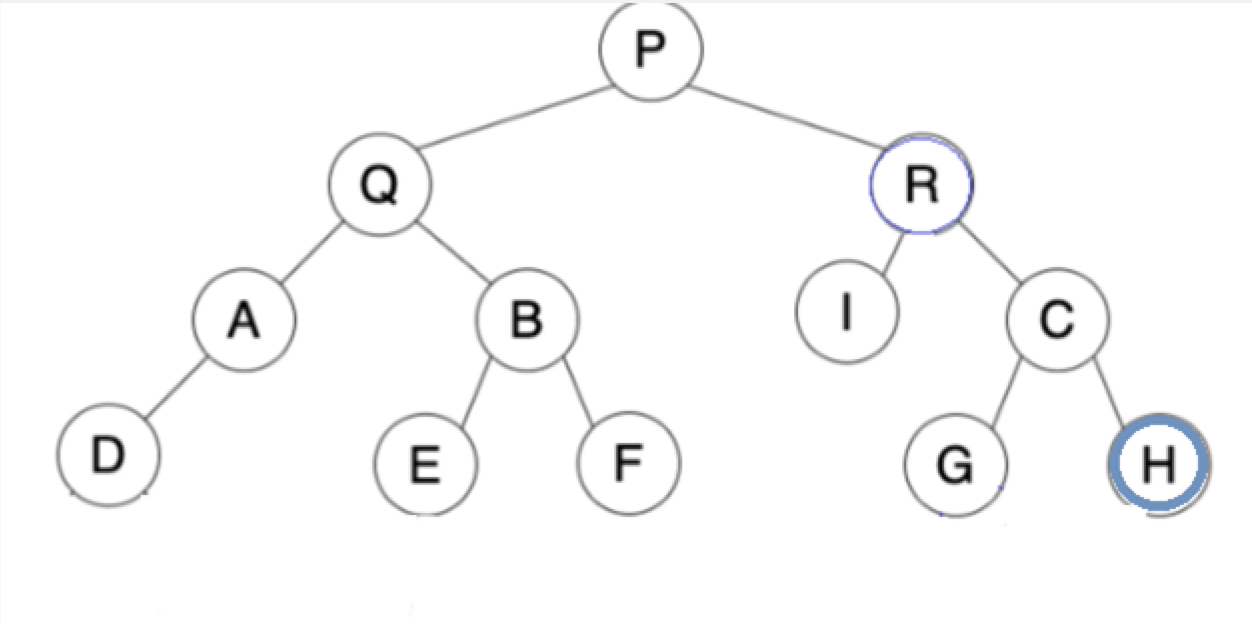


Elements Of Tree

Node: It represents a termination point in a tree.
Root: A tree's topmost node.
Parent: Each node (apart from the root) in a tree that has at least one sub-node of its own is called a parent node.
Child: A node that straightway came from a parent node when moving away from the root is the child node.
Leaf Node: These are external nodes. They are the nodes that have no child.
Internal Node: As the name suggests, these are inner nodes with at least one child.
Depth of a Tree: The number of edges from the tree's node to the root is.
Height of a Tree: It is the number of edges from the node to the deepest leaf. The tree height is also considered the root height.

Binary Tree

A binary tree is a tree-type non-linear data structure with a maximum of two children for each parent. Every node in a binary tree has a left and right reference along with the data element. The node at the top of the hierarchy of a tree is called the root node. The nodes that hold other sub-nodes are the parent nodes.



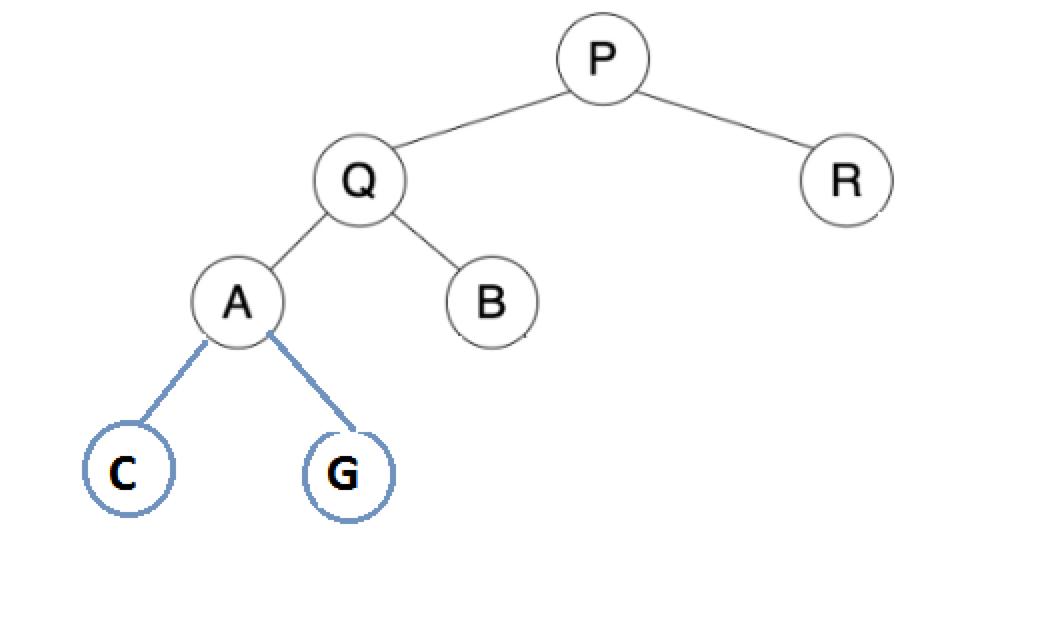
Benefits Of Binary Tree

- The search operation in a binary tree is faster as compared to other trees
- Only two traversals are enough to provide the elements in sorted order
- It is easy to pick up the maximum and minimum elements
- Graph traversal also uses binary trees
- Converting different postfix and prefix expressions are possible using binary trees

Types Of Binary Tree

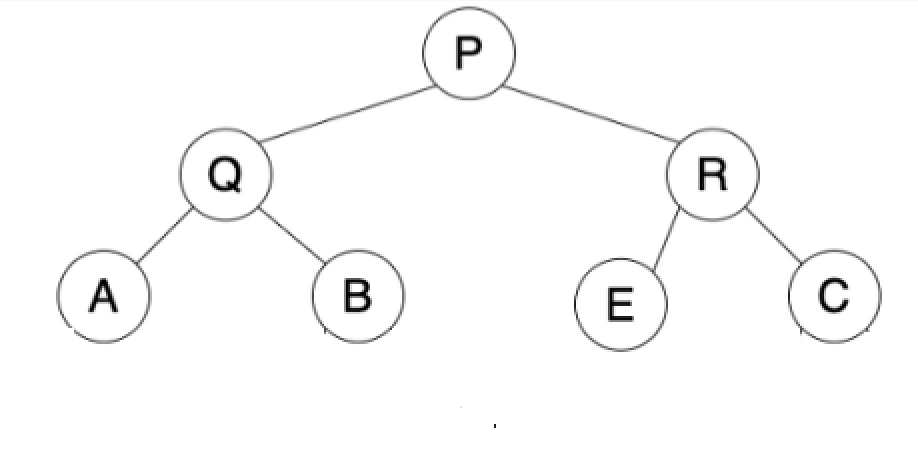
1. Full Binary Tree.

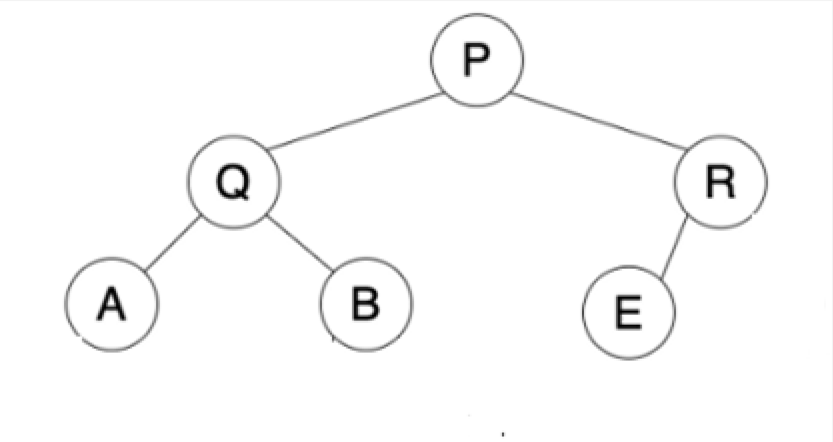
It is a special kind of a binary tree that has either zero children or two children. It means that all the nodes in that binary tree should either have two child nodes of its parent node or the parent node is itself the leaf node or the external node.



2. Complete Binary Tree.

A complete binary tree is another specific type of binary tree where all the tree levels are filled entirely with nodes, except the lowest level of the tree. Also, in the last or the lowest level of this binary tree, every node should possibly reside on the left side. Here is the structure of a complete binary tree:



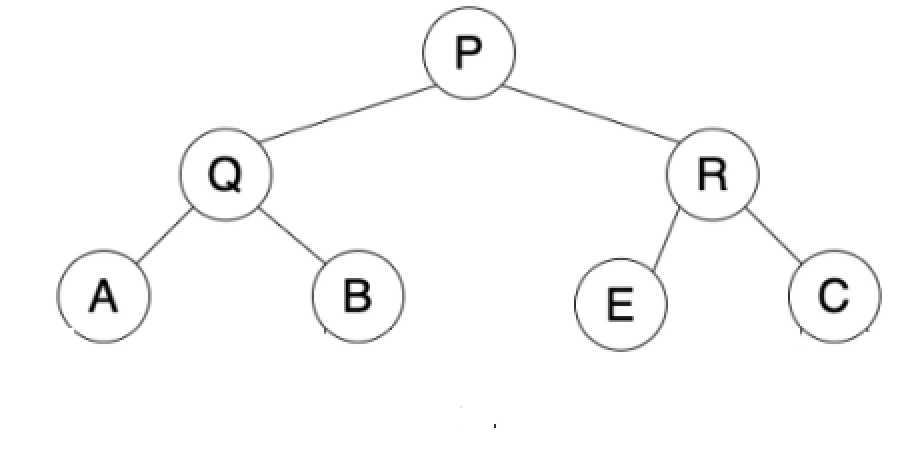


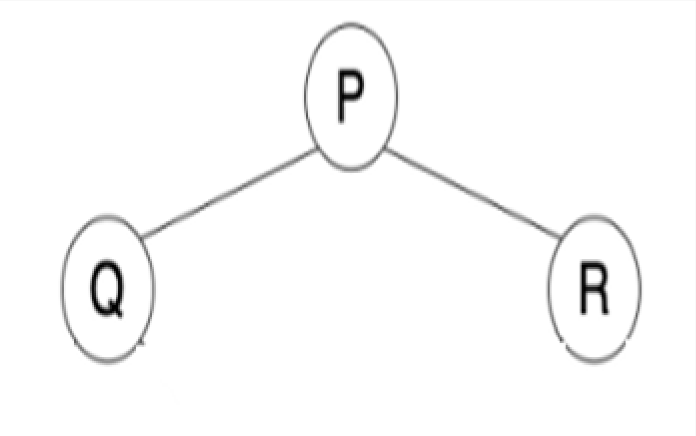
Types Of Binary Tree

OBJECT TECHNOLOGIES

3. Perfect Binary Tree.

A binary tree is said to be 'perfect' if all the internal nodes have strictly two children, and every external or leaf node is at the same level or same depth within a tree. A perfect binary tree having height 'h' has 2h – 1 node. Here is the structure of a perfect binary tree:





Binary Tree Representation

Binary Tree Representation

A node of a binary tree is represented by a structure containing a data part and two pointers to other structures of the same type.